Machine Learning · 3rd-Year Software Engineering

Chapter 2 — Data Preparation & Exploratory Data Analysis

Session 1 of 2 · Data types, descriptive statistics, missing values, outliers

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On real data, one naive cleaning rule can delete almost 80% of the rows

891

passengers in the Titanic dataset (seaborn-data copy)

177

missing ages (19.9%)

688

missing deck values (77.2%)

182

rows left after dropping every row with a missing age, deck or embarked

Before any model: what would you do with these gaps? A · drop rows B · fill the values C · drop columns

By the end of this session, you can do four things

Observable outcomes — each one is practised and checked today

1. Classify a feature as continuous, discrete, nominal or ordinal
2. Compute mean, variance and standard deviation by hand, and say whether a tool divides by n or n − 1
3. Choose and justify a treatment for missing values
4. Detect outliers with the IQR rule and decide what to do with them

Session 2

- Encoding categories
- Scaling numbers
- Correlation & EDA
- Data leakage

Recall from Chapter 1: why is the test set sacred?

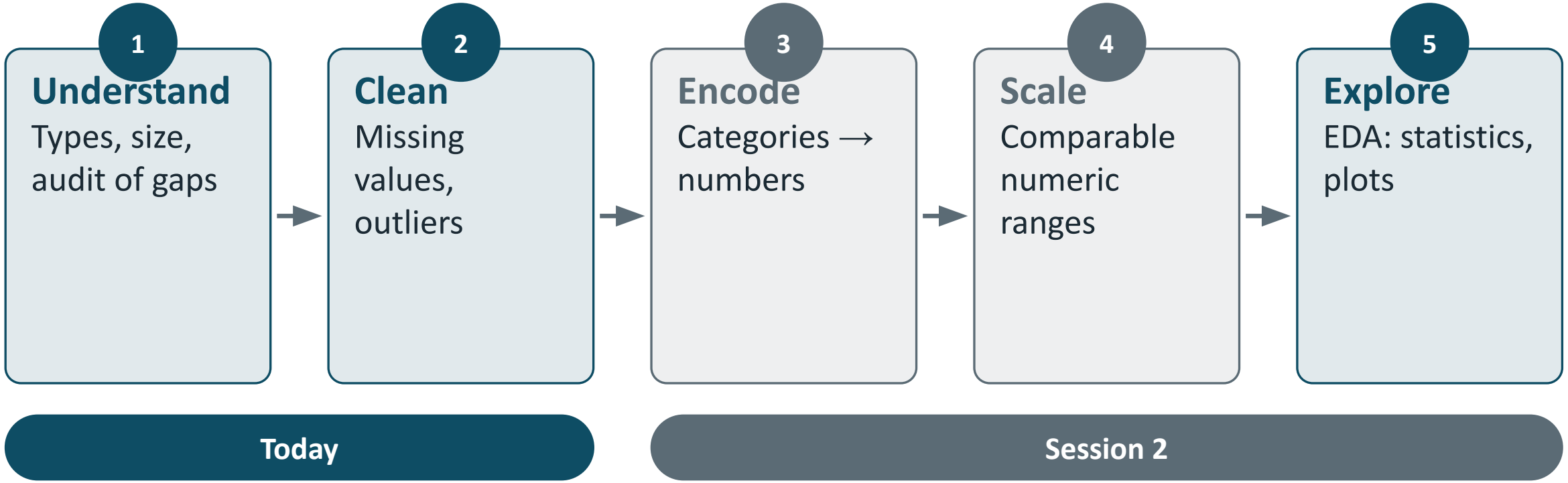
RECALL · 1 min · individual

Why must the test set stay untouched until the final evaluation? Write one sentence.

Hint: what is the test score supposed to estimate?

Think alone for 1 minute, then two volunteers answer. No notes.

Data preparation is a pipeline of five steps



Rule of the chapter: split first, fit every step on the training set only, then transform both sets

PART 1

Know your data types

The type of a feature decides how we clean, encode and scale it

Features are numerical or categorical — each family has two kinds

Numerical · continuous
Any value in a range

Titanic: age, fare

Numerical · discrete
Counts (whole numbers)

Titanic: sibsp, parch (relatives aboard)

Categorical · nominal
Labels with no order

Titanic: sex, embarked (C / Q / S)

Categorical · ordinal
Labels with a meaningful order

Titanic: class (1st / 2nd / 3rd); low < medium < high

Models compute on numbers: categorical features must be converted, and how we convert them encodes an assumption

Your turn: classify six features

ACTIVITY · 2 min · individual

What type is each feature?

1 · T-shirt size (S / M / L)	2 · Country	3 · Satisfaction (1–5 stars)
4 · Blood type	5 · sibsp (relatives aboard)	6 · fare (ticket price)

Show fingers for each item: 1 = continuous · 2 = discrete · 3 = nominal · 4 = ordinal

Correction: a digit does not make a feature numerical

Feature	Type	Why
T-shirt size	Ordinal	S < M < L
Country	Nominal	No natural order
Satisfaction (stars)	**Ordinal**	**Ordered; equal gaps not guaranteed**
Blood type	Nominal	No natural order
sibsp	Numerical · discrete	A count
fare	Numerical · continuous	Any amount in a range

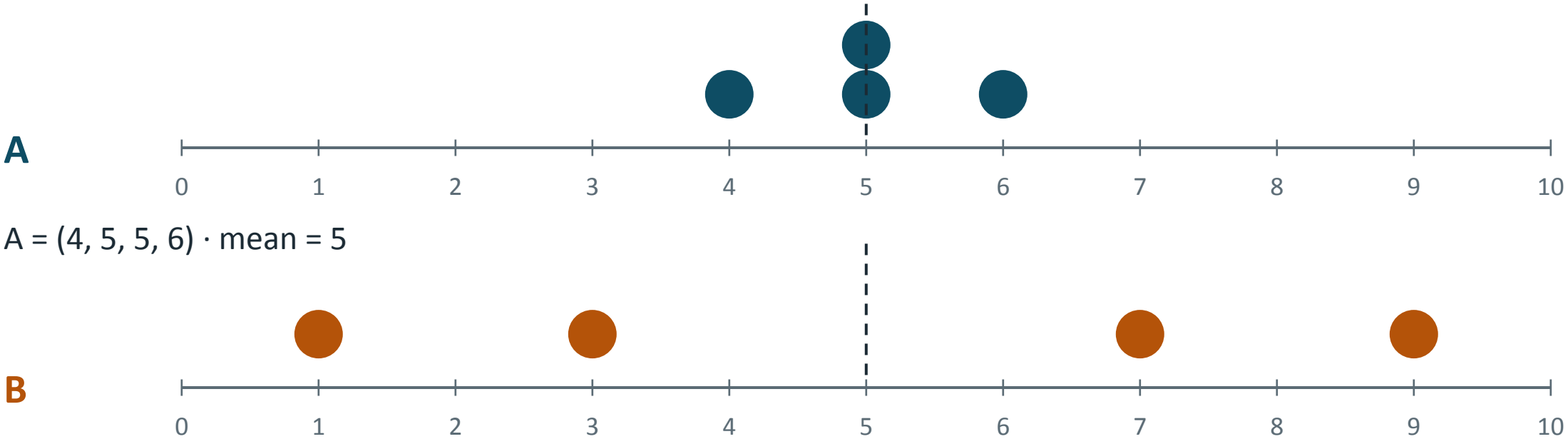
Stars can be treated as numbers only if you assume the gap 1→2 equals the gap 4→5 — state that assumption explicitly

PART 2

Describe a numerical feature

Centre and spread: mean, variance, standard deviation

Two datasets can share the same mean and still be very different



A = (4, 5, 5, 6) · mean = 5

B = (1, 3, 7, 9) · mean = 5

Question: which single number would capture the difference between A and B?

Mean measures the centre; variance and standard deviation measure the spread

Mean

$$\bar{x} = (1/n) \cdot \Sigma\, x_i$$

The balance point of the values

Variance (population)

$$\sigma^2 = (1/n) \cdot \Sigma\, (x_i - \bar{x})^2$$

Average squared distance to the mean

Standard deviation

$$\sigma = \sqrt{\sigma^2}$$

Back in the unit of the data (years, dinars…)

Why square?

Raw deviations always sum to 0: positives cancel negatives.

Squaring makes them positive and weighs large deviations more.

Worked example: mean and standard deviation in three rows

Data: x = (2, 4, 4, 4, 5, 5, 7, 9), n = 8. Step 1: $\bar{x} = 40 / 8 = \mathbf{5}$

									Σ
$\mathbf{x_i}$	2	4	4	4	5	5	7	9	**40**
$\mathbf{x_i - \bar{x}}$	−3	−1	−1	−1	0	0	2	4	**0**
$\mathbf{(x_i - \bar{x})^2}$	9	1	1	1	0	0	4	16	**32**

Step 2: $\sigma^2 = 32 / 8 = \mathbf{4}$ Step 3: $\sigma = \sqrt{4} = \mathbf{2}$

Check: the deviations sum to 0 ✓ (also verified with NumPy: np.std(x) = 2.0)

Divide by n or by n − 1? Your library decides silently

Same data x (n = 8)	Divides by	Result
Population formula σ	n	2.000
Sample formula s (Bessel)	n − 1	2.138
np.std(x)	n	2.000
pd.Series(x).std()	n − 1	2.138
StandardScaler().fit(X).scale_	n	2.000

Why n − 1?

The mean is itself estimated from the same sample; dividing by n − 1 corrects the resulting underestimation of the variance.

Rule: know your tool's ddof.

Verified by execution with NumPy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1 — re-check after major upgrades

Your turn: measure the spread of dataset B

MINI-CALCULATION · 3 min · individual, paper

B = (1, 3, 7, 9). Compute the mean and the population standard deviation σ.

Reuse the three-row table: $x_i \cdot x_i - \bar{x} \cdot (x_i - \bar{x})^2$

Leave σ as a square root if you have no calculator

Then compare with your neighbour for 30 seconds before the correction.

Correction: same mean, but B is about 4.5 times more spread out than A

					Σ
x_i	1	3	7	9	**20**
$x_i - \bar{x}$	−4	−2	2	4	**0**
$(x_i - \bar{x})^2$	16	4	4	16	**40**

$\bar{x} = 20 / 4 = \mathbf{5}$ $\sigma^2 = 40 / 4 = \mathbf{10}$ $\sigma = \sqrt{10} \approx \mathbf{3.16}$

For A = (4, 5, 5, 6): $\sigma = \sqrt{0.5} \approx 0.71 \rightarrow$ ratio ≈ 4.5. This is the number that answers the question of the dot plot.

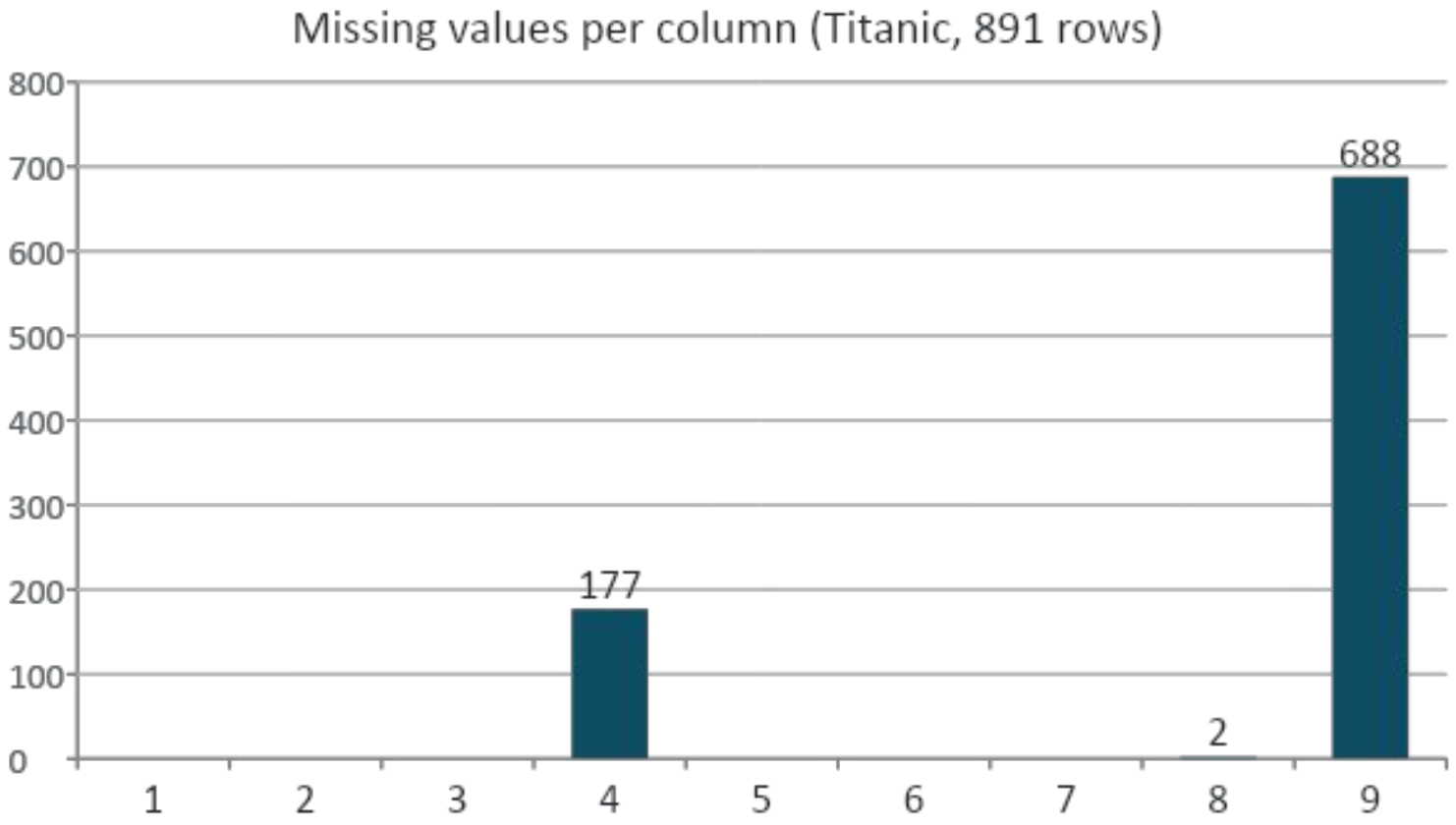
Report the spread together with the mean: a mean alone hides how typical it is

PART 3

Handle missing values

Audit first, then drop or impute — and justify the choice

Missing values are uneven: one column is mostly empty, another barely touched



As a share of 891 rows

deck: 77.2%

age: 19.9%

embarked: 0.2% (2 rows)

Audit line: df.isna().sum()

Three options for a gap — each has a cost

Drop the rows

When: very few rows affected

Cost: lost data; bias if the missing rows differ from the others

Titanic: dropna on 3 columns keeps 182 / 891

Drop the column

When: the column is mostly empty

Cost: any signal it carried is lost

Titanic: deck, 77.2% missing

Impute (fill)

When: moderate missingness

Numerical: median or mean

Categorical: most frequent or a "Missing" category

Cost: distorts the distribution

Imputation = replacing a missing value by an estimated one. The fill value is computed from data, so it is learned

Predict: which fill value is more plausible?

PREDICTION · 1 min · vote

Synthetic ticket prices: 7, 8, 8, 9, 10, ?, 250. One value is missing. Which fill value describes a typical passenger better?

A · the mean of the six known values

B · the median of the six known values

Raise one finger for A, two for B. Decide before computing.

One extreme fare drags the mean; the median stays with typical passengers

48.67

Mean = 292 / 6 — larger than 5 of the 6 known fares

8.5

Median = (8 + 9) / 2 — the middle of the sorted values

Why

The mean uses the size of 250; the median only uses its position (it is just 'the largest').

Rule of thumb

Skewed numerical column → median. Roughly symmetric → mean is fine.

Imputation has limits you must know before using it

Artificial spike

Filling 177 ages with one value makes 19.9% of rows identical; a model may latch onto it.

Missingness can be information

If being missing carries signal, add an indicator column, e.g. age_was_missing (0/1).

Categorical gaps

Fill with the most frequent category, or create an explicit "Missing" category.

The fill value is learned

Compute it on the training split only — never on the full dataset (session 2).

Your turn: decide for each Titanic column

ACTIVITY · 3 min · pairs

For each column: drop rows, drop the column, or impute (with which value)? Justify in one sentence.

age · numerical · 19.9% missing

deck · categorical · 77.2% missing

embarked · nominal · 2 rows missing (S 644 · C 168 · Q 77)

Discuss with your neighbour; three pairs will present one decision each.

Correction: targeted decisions keep all 891 passengers

Column	Decision	Justification
age	Impute the median (+ optional age_was_missing)	80% of values present; median robust to skew
deck	Drop the column (or keep "Missing" as a category to test it)	77.2% empty: imputing would mostly invent data
embarked	Impute the most frequent value "S"	Only 2 rows; S = 644 / 889 known (72.4%)

Back to the opening vote: dropping every incomplete row kept 182 passengers; column-by-column decisions keep 891

PART 4

Detect and decide on outliers

An extreme value is a question, not automatically an error

Titanic fares: a few very high values pull the mean far above the median

14.45

median fare

32.20

mean fare

512.33

maximum fare (n = 891)

Error
Impossible value, e.g. age = 250 years → correct or remove

Rare but real
Valid extreme value, e.g. a very expensive ticket → usually keep

Other population
A different group mixed in → study it separately

Two detection rules: the IQR fences and the z-score

Quartiles and interquartile range

$$\mathrm{IQR} = Q_3 - Q_1$$

Q1 = 25th percentile, Q3 = 75th percentile: the middle 50% of the data

IQR rule (Tukey's fences)

$$x < Q_1 - 1.5 \cdot \mathrm{IQR} \quad \text{or} \quad x > Q_3 + 1.5 \cdot \mathrm{IQR}$$

Uses quartiles, so the outliers barely move the fences

z-score rule

$$|z| > 3 \quad \text{with} \quad z = (x - \bar{x}) \,/\, \sigma$$

Best for roughly bell-shaped data

Limits

1.5 and 3 are conventions, not laws.

An extreme value inflates σ itself, which can hide it from the z-score rule.

Worked example: the IQR rule flags 250 and nothing else

Synthetic data: 7, 8, 8, 9, 10, 12, 250 (n = 7)

Step	Value
Q1 (25th percentile)	8
Q3 (75th percentile)	11
IQR = Q3 − Q1	3
Lower fence 8 − 1.5×3	3.5
Upper fence 11 + 1.5×3	**15.5**

Conventions differ

NumPy and pandas default (linear interpolation) give Q3 = 11.

Other conventions give Q3 = 10 or 12 → upper fence 13 or 18.

Same verdict here; borderline points can change.

Verdict: 250 > 15.5 → flagged; every other value lies inside [3.5, 15.5]

Once detected, choose among four actions — using domain knowledge

Keep

Real and relevant value

Prefer robust tools (median, IQR)

Cap (winsorize)

Replace by the fence or a high percentile

Keeps the row, limits its influence

Remove

Only a proven error

Document what and why

Transform

e.g. log for right-skewed positive data

Titanic fare has zeros → log(1 + x)

Detection is automatic; the decision is not. Write the decision down so it can be reproduced

Exit check: true or false?

EXIT CHECK · 3 min · individual

Thumb up = true, thumb down = false — vote item by item on the count of 3.

1 · An outlier is always a data-entry error.

2 · For a right-skewed column, the median is less affected by extreme values than the mean.

3 · On the same data, pandas .std() and np.std() return the same value by default.

4 · Dropping every row that has a missing value is always a safe default.

Exit check correction

#	Answer	Why
1	False	It can be a rare but real value or another population; check before acting
2	True	The median depends on positions, not on how extreme the largest value is
3	False	pandas divides by n − 1 (ddof = 1), NumPy by n (ddof = 0): 2.138 vs 2.000
4	False	On Titanic it kept 182 of 891 rows and may bias what remains

What to remember from session 1

- The type decides the treatment: a digit is not necessarily numerical
- Report a spread with every mean; know whether your tool divides by n or n − 1
- Missing values: audit, then decide column by column — drop, drop the column, or impute
- Skewed data: the median resists extreme values better than the mean
- Outliers: detect with a rule (IQR), decide with domain knowledge, document it

Session 2

- Encoding categories
- Scaling numbers
- Correlation & EDA on Titanic
- Data leakage: why every fill value and scale must come from the training split

References and data used in this session

- Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley. — box plots and 1.5 × IQR fences
- McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly. Open-access web edition: wesmckinney.com/book
- James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). An Introduction to Statistical Learning with Applications in Python. Springer. doi:10.1007/978-3-031-38747-0
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830
- Data: Titanic dataset as distributed in the seaborn-data repository (github.com/mwaskom/seaborn-data), titanic.csv, 891 rows × 15 columns

All figures on these slides were recomputed by executing Python (NumPy 2.5.3, pandas 3.0.5, scikit-learn 1.9.1) on that file. Synthetic examples are labelled as such.